Homebrew DMR trunking protocol

Version 0.6

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1. Example UDP message format

Byte	Value	Comment
0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	Opcode 0000 0000 - 0111 1111	MSB 1 indicates internal (1) or external (0) MSB 2 indicates direction (0 to network, 1 from network) MSB 3 – 8: Operation code
5	Unix timestamp (signed 64 bit LE)	Unix time (seconds, UTC)
6		
7		
8		
9		
10		
11		
12		
13		UUID (16 bytes)
17		
21		
25		
29	Payload size	(number of addresses or byte size)
30	First value	Byte 0 (BE)
31	First value	Byte 1
32	First value	Byte 2
33	Second value	Byte 0
...	...	continued

2.1. Network Opcodes

Binary	Hex	Usage
0000 0000	0x00	Reserved
0100 0000	0x40	Reserved
0000 0001	0x01	Registration (to network)
0100 0001	0x41	Registration confirmation (from network)
0000 0010	0x02	De-registration (to network)
0100 0010	0x42	De-registration (from network)
0000 0011	0x03	Group subscription (to network)
0100 0011	0x43	Group subscription confirmation (from network)
0000 0100	0x04	Group un-subscription (to network)
0100 0100	0x44	Group un-subscription confirmation (to network)
0000 0101	0x05	UDT transfer (to network)
0100 0101	0x45	UDT transfer (from network)
0000 0110	0x06	UDT accept (to network)
0100 0110	0x46	UDT accept (from network)
0000 0111	0x07	Private call setup (to network)
0100 0111	0x47	Private call setup (from network)
0000 1000	0x08	Private call accept (to network)
0100 1000	0x48	Private call accept (from network)
0000 1001	0x09	Private call reject (to network)
0100 1001	0x49	Private call reject (from network)
0000 1010	0x0A	Status transfer (to network)
0100 1010	0x4A	Status transfer (from network)

All adresses (MS, group, system) are transferred in the BigEndian format.
Unix timestamps (where required) are transferred in the LittleEndian format.
UUID (where required) is transferred in the BigEndian format.
16 bytes UUID is appended to all DMRD data packets.
Message size must not exceed 256 bytes.

Header signature changes							
RPTL	RPTK	RPTC	RPTCL	RPTACK	MSTNAK	RPTPING	MSTPONG
DTCL	DTCK	DTCC	DTCCL	DTCACK	DTCNAK	DTCPING	DTCPONG

2.2. Internal Opcodes

Binary	Hex	Usage
1000 0000	0x80	Reserved
1100 0000	0xC0	Reserved
1100 0001	0xC1	Master login information
1100 0010	0xC2	Master close information
1000 0001	0x81	Channel enable / disable
1000 0010	0x82	Reverse channel Cease Transmission
1000 0011	0x83	Reverse channel Request Cease Transmission
1000 0100	0x84	Reverse channel Power Increase One Step
1000 0101	0x85	Reverse channel Power Decrease One Step
1000 0110	0x86	Reverse channel Maximum Power
1000 0111	0x87	Reverse channel Minimum Power

3. Registration message (to network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0000 0001	Opcode: Registration request
5	Unix timestamp (signed 64 bit LE)	Unix time (UTC)
6		
7		
8		
9		
10		
11		
12		
13	0000 0001	Size (no. of addresses)
14	MS address	Byte 0
15	MS address	Byte 1
16	MS address	Byte 2

4. Registration confirmation (from network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0100 0001	Opcode: Registration confirmation
5	Unix timestamp (signed 64 bit LE)	Unix time (UTC)
6		
7		
8		
9		
10		
11		
12		
13	Status bitfield (LSB first, up to 8 MS addresses)	1 - Accepted registration 0 - Denied registration
14	0000 0001	Size (no. of addresses)
15	MS address	Byte 0
16	MS address	Byte 1
17	MS address	Byte 2

5. De-registration message (to network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0000 0010	Opcode: De-registration
5	Unix timestamp (signed 64 bit LE)	Unix time (UTC)
6		
7		
8		
9		
10		
11		
12		
13	0000 0001	Size (no. of addresses)
14	MS address	Byte 0
15	MS address	Byte 1
16	MS address	Byte 2

6. De-registration message (from network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0100 0010	Opcode: De-registration
5	Unix timestamp (signed 64 bit LE)	Unix time (UTC)
6		
7		
8		
9		
10		
11		
12		
13	0000 0001	Size (no. of addresses)
14	MS address	Byte 0
15	MS address	Byte 1
16	MS address	Byte 2

7. Group subscription (to/from network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0000 0011 0100 0011	Opcode: 1. Group subscription 2. Subscription confirmation
5	Size (8 bit unsigned)	Talkgroup size (no. of addresses) 1 – 64, max. 64
6	Group 1 address	Byte 0
7	Group 1 address	Byte 1
8	Group 1 address	Byte 2
9	Group 2 address	Byte 0
19	Etc.	continued

Subscription confirmation contains the same group IDs that were asked in group subscription, if core was able to subscribe them. Groups that failed to subscribe will be set to zero in the packet payload.

8. Group un-subscription (to/from network)

0	‘D’	Magic header
1	‘M’	
2	‘R’	
3	‘T’	
4	0000 0100 0100 0100	Opcode: 1. Group un-subscription 2. Un-subscription confirmation
5	Size (8 bit unsigned)	Talkgroup size (no. of addresses) 1 – 64, max 64
6	Group 1 address	Byte 0
7	Group 1 address	Byte 1
8	Group 1 address	Byte 2
9	Group 2 address	Byte 0
10	Etc.	continued

9. UDT transfer: SMS, location, number etc.

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 0101 0100 0101				Opcode: 1. UDT transfer (to network) 2. UDT transfer (from network)
5					UUID (16 bytes)
9					
13					
17					
21	*Type* 0000 0011 – private SMS (ISO-7) 1000 0100 – group SMS (ISO-8) - only one supported now 1000 0111 – group SMS (UTF-16) etc.				MSB 1: private / group flag MSB 2: acknowledge flag LSB 1-4: format Table 9.51: UDT Format information element content
22	Payload size (8 bit unsigned)				Byte size : 1 - 46 (max 46 char ISO-8)
23	Source address				Byte 0
24	Source address				Byte 1
25	Source address				Byte 2
26	Destination address				Byte 0
27	Destination address				Byte 1
28	Destination address				Byte 2
29	Payload				Byte 0
30	Payload				Byte 1
31	Etc.				continued

10. UDT accept message

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 0110 0100 0110				Opcode: 1. UDT accept (to network) 2. UDT accept (from network)
5					UUID (16 bytes) Should match UDT transfer UUID
9					
13					
17					
21	Source address				Byte 0 (BE)
22	Source address				Byte 1
23	Source address				Byte 2
24	Destination address				Byte 0
25	Destination address				Byte 1
26	Destination address				Byte 2

11. Private call setup

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 0111 0100 0111				Opcode: 1. Private call setup (to network) 2. Private call setup (from network)
5					UUID (16 bytes)
9					
13					
17					
21	Service options				7 bit, MSB 1 reserved, set to 0
22	Source address				Byte 0 (BE)
23	Source address				Byte 1
24	Source address				Byte 2
25	Destination address				Byte 0
26	Destination address				Byte 1
27	Destination address				Byte 2

12. Private call accept

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 1000 0100 1000				Opcode: 1. Private call accept (to network) 2. Private call accept (from network)
5					UUID (16 bytes) Should match call setup UUID
9					
13					
17					
21	Source address				Byte 0 (BE)
22	Source address				Byte 1
23	Source address				Byte 2
24	Destination address				Byte 0
25	Destination address				Byte 1
26	Destination address				Byte 2

13. Private call reject

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 1001 0100 1001				Opcode: 1. Private call reject (to network) 2. Private call reject (from network)
5					UUID (16 bytes) Should match call setup UUID
9					
13					
17					
21	Source address				Byte 0 (BE)
22	Source address				Byte 1
23	Source address				Byte 2
24	Destination address				Byte 0
25	Destination address				Byte 1
26	Destination address				Byte 2

14. Status delivery service

0	‘D’				Magic header
1	‘M’				
2	‘R’				
3	‘T’				
4	0000 1010 0100 1010				Opcode: 1. Status call (to network) 2. Status call (from network)
5					UUID (16 bytes)
9					
13					
17					
21	*Status* 0000 0001 – private Status 1000 0001 – group Status				MSB 1: group / private call flag MSB 2-7: status (128 values) Special values defined at 6.6.6.0 Status Call Service - Introduction
22	Source address				Byte 0 (BE)
23	Source address				Byte 1
24	Source address				Byte 2
25	Destination address				Byte 0
26	Destination address				Byte 1
27	Destination address				Byte 2